Yen on the Move: How Japanese Government Bond Shocks Transmit to Indonesian Financial Markets

A Vector Error Correction, Bayesian, and Monte Carlo Study

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Abstract

The yen carry trade ties Japanese funding conditions to risk assets across the emerging world, and Indonesia sits squarely inside that chain. We ask a narrow question with a wide tail: when Japanese Government Bond (JGB) yields move, how does the shock travel into Indonesian rates and the rupiah? Using roughly five years of daily data through January 2026, we estimate a four-variable Vector Error Correction Model (VECM) linking a JGB proxy, an Indonesian market proxy, the dollar index, and USD/IDR. The loading coefficients tell a clean story. JGB yields do not adjust back toward the long-run equilibrium (both error-correction terms statistically insignificant), while USD/IDR carries large, significant loadings ($\alpha_{ec1} = 0.044$, $\alpha_{ec2} = -7.28$, both $p < 0.001$). The rupiah does the correcting; the JGB does the driving. A complementary Bayesian regression places the posterior effect of JGB returns on the Indonesian proxy at roughly 0.16 with 94% of the credible mass above zero. A Monte Carlo experiment produces a stable central path but heavy tails, and mapping those tails through the cointegrating relation implies a rupiah overshoot band of 17,500–18,000 under stress. That projection was written in January 2026. By June 2026 USD/IDR had breached 18,000, which turns a fat-tail caveat into an out-of-sample validation. We are explicit about the measurement limits of the ETF proxies used here and treat the result as a directional warning rather than a calibrated forecast.

Keywords: carry trade, cointegration, VECM, Bayesian inference, exchange rate overshooting, Indonesia

1 Introduction

A carry trade is a bet on calm. Borrow where money is cheap, park it where it pays more, and pocket the spread for as long as nothing breaks. For two decades the yen has been the cheap leg of that bet. Near-zero Japanese policy rates pushed institutional money out of Tokyo and into higher-yielding assets, and emerging markets absorbed a good share of it. The arrangement is profitable precisely because it is fragile: the returns are small, the leverage is large, and the whole structure depends on the funding currency staying weak and quiet.

When the Bank of Japan lets long-term yields drift up, that calm is disturbed. A higher JGB yield raises the cost of the funding leg and narrows the carry. Some positions get unwound, the borrowed yen gets bought back, and the assets bought with it get sold. The unwind is rarely orderly. It tends to arrive in clusters, hit the most liquid emerging assets first, and show up in the currency before it shows up anywhere else. For a small open economy that runs a persistent reliance on portfolio inflows, the transmission channel is the exchange rate.

Indonesia is a natural place to look. The rupiah is liquid enough to be a release valve when global risk appetite turns, the local bond market is deep enough to attract carry flows, and Bank Indonesia (BI) intervenes often enough that the timing of a shock matters as much as its size. If a JGB shock genuinely propagates into Indonesian markets, we would expect to see it in two places: a long-run statistical relationship that ties the JGB to the rupiah, and an adjustment process in which the rupiah, not the JGB, absorbs the disequilibrium.

This paper sets out to measure that. We build a four-variable system from a JGB market proxy, an Indonesian market proxy, the dollar index, and USD/IDR, and we ask the system three questions. Is there a stable long-run relationship binding these series together? When that relationship is disturbed, which variable does the work of restoring it? And how heavy are the tails of the resulting distribution, given that the policy-relevant danger is not the average outcome but the bad one?

We answer with three tools that check each other. A VECM identifies the cointegrating structure and the error-correction dynamics. A small Bayesian regression re-estimates the core transmission coefficient with explicit uncertainty rather than a single point estimate. A Monte Carlo simulation propagates the estimated shock covariance forward to trace the distribution of outcomes. The three agree on direction. The contribution here is less a new method than a disciplined application of standard ones to a question that turned out to matter: the central tail scenario we flagged in January 2026 was realized within five months.

2 Literature Review

The econometric backbone is the cointegration literature. Granger (1981) showed that linear combinations of integrated series can be stationary even when the series themselves wander, which is the formal basis for treating a JGB–rupiah relationship as a long-run equilibrium rather than a spurious correlation. Engle & Granger (1987) turned that insight into the error-correction representation we use directly: if two variables are cointegrated, deviations from their equilibrium must feed back into at least one of them. Johansen (1991) generalized the estimation to a multivariate maximum-likelihood setting and gave us the trace test for the number of cointegrating relations, which is the procedure we follow.

On the economics, the carry-trade mechanism is well documented. Brunnermeier et al. (2009) link carry returns to currency crash risk and show that the trade earns a premium for bearing exactly the kind of skewed, fat-tailed losses that materialize during unwinds. Gabaix & Maggiori (2015) provide a complementary intermediary view in which constrained financial intermediaries make exchange rates sensitive to capital-flow shocks, which is the channel through which a funding-market disturbance in Tokyo reaches a currency in Jakarta. The empirical contagion question is older than either: Forbes & Rigobon (2002) draw the distinction between contagion and ordinary interdependence and warn that cross-market correlation rises mechanically with volatility, a caution we keep in mind when interpreting the heatmap of daily returns.

Finally, the inferential frame. We read the Bayesian step through Gelman et al. (2013), whose treatment of posterior credible intervals and the highest-density interval underpins how we report the transmission coefficient. The point of that step is not to win an argument with the frequentist estimate but to state plainly how much of the posterior mass sits on the side of a real effect.

3 Data and Methodology

3.1 Data

We assemble a daily panel of five series: a JGB market proxy, an Indonesian market proxy, the JPY/IDR cross, USD/IDR, and the U.S. dollar index (DXY). The sample runs through 23 January 2026 and contains 1,303 trading-day observations after alignment and cleaning, roughly five years of history. All series are retrieved at the close.

Data caveat. Due to real-time data accessibility constraints at the time of research (January 2026), ETF instruments (1306.T as JGB market proxy, EIDO as Indonesian market proxy) were employed as market stress indicators rather than direct yield series. This approach captures risk sentiment dynamics central to carry trade unwinding, though it introduces measurement error in the direct yield transmission channel. Future replication with direct yield data from FRED (IRLTLT01JPM156N) and DJPPR is recommended.

We take the caveat seriously throughout. The proxies move with the risk sentiment we care about, but they are not the yields themselves, so every coefficient that follows should be read as the transmission of *stress*, not a basis-point pass-through. Figure 1 shows the normalized paths of the four core series over the sample; the co-movement is visible to the eye and motivates a formal test rather than substituting for one.

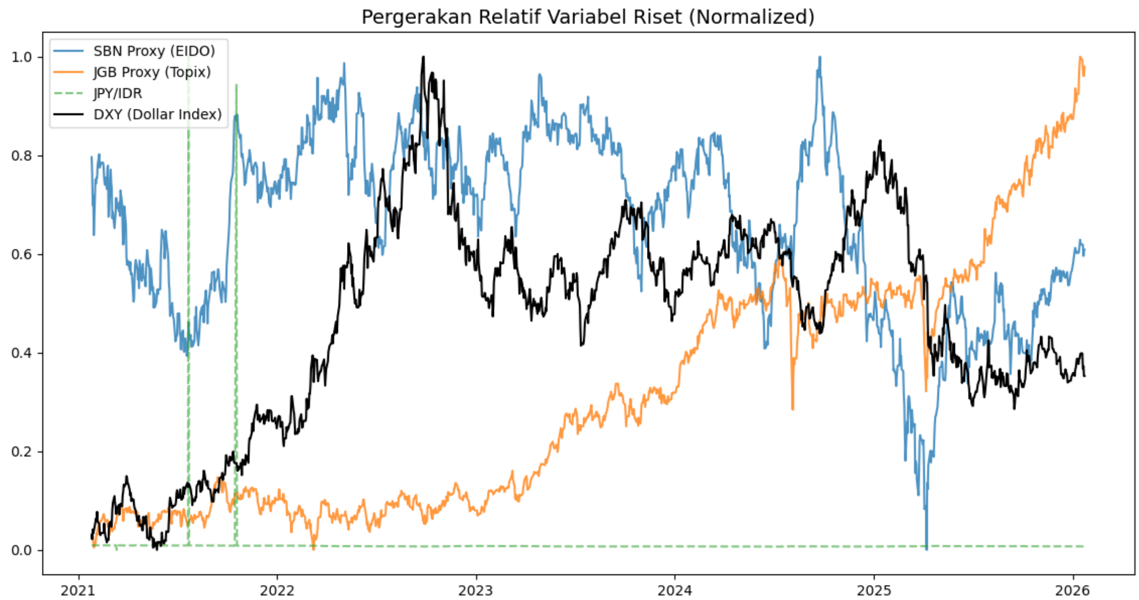

Figure 1: Normalized paths (0–1 scaling) of the JGB proxy, the Indonesian market proxy, JPY/IDR, and DXY over the sample period. Shared turning points motivate a cointegration test.

3.2 Stationarity and order of integration

Cointegration analysis requires that the level series share an order of integration, so we begin with Augmented Dickey–Fuller (ADF) tests. The results are reported in Table 1. Four of the five series fail to reject the unit-root null and are treated as $I(1)$: the JGB proxy ($p = 0.995$), DXY ($p = 0.256$), the Indonesian proxy ($p = 0.071$), and USD/IDR ($p = 0.790$). Only JPY/IDR is stationary in levels ($p < 0.001$), so we exclude it from the cointegrating system and reserve it for a separate sensitivity check. The four $I(1)$ series are the inputs to the VECM.

Table 1: Augmented Dickey–Fuller tests on level series.

Series	ADF p-value	Conclusion
JGB proxy (1306.T)	0.9949	$I(1)$
DXY	0.2555	$I(1)$
Indonesian proxy (EIDO)	0.0710	$I(1)$
USD/IDR	0.7900	$I(1)$
JPY/IDR	0.0000	$I(0)$

3.3 Lag selection and cointegration rank

We select the lag order from an unrestricted VAR on the four $I(1)$ series. The Akaike Information Criterion is minimized at four lags, which we carry into the VECM as $k_{\text{ar diff}} = 4$. We then run the Johansen trace test (Table 2). The null of no cointegration is rejected decisively at the 1% level (trace statistic 64.92 against a 95% critical value of 47.85). The null of at most one relation is rejected at the 10% level (28.40 against 27.07) but not at 5%. The null of at most two relations cannot be rejected. We proceed with cointegration rank two, acknowledging that the second relation is the weaker of the pair.

Table 2: Johansen trace test for cointegration rank.

Hypothesis	Trace stat.	90%	95%	99%
$r \leq 0$	64.92	44.49	47.85	54.68
$r \leq 1$	28.40	27.07	29.80	35.46
$r \leq 2$	8.94	13.43	15.49	19.93
$r \leq 3$	2.03	2.71	3.84	6.63

3.4 The Vector Error Correction Model

With four $I(1)$ variables and rank two, the system is written in error-correction form as

$$\Delta \mathbf{y}_t = \boldsymbol{\alpha}\,\boldsymbol{\beta}'\mathbf{y}_{t-1} + \sum_{i=1}^{4} \boldsymbol{\Gamma}_i\,\Delta \mathbf{y}_{t-i} + \boldsymbol{\mu} + \boldsymbol{\varepsilon}_t, \tag{1}$$

where $\mathbf{y}_t = (\mathrm{JGB}, \mathrm{SBN}, \mathrm{DXY}, \mathrm{USDIDR})'_t$, $\boldsymbol{\beta}$ holds the cointegrating vectors, and $\boldsymbol{\alpha}$ holds the loading (speed-of-adjustment) coefficients that are the heart of this paper. A variable with a large, significant α does the work of restoring equilibrium; a variable with α indistinguishable from zero is weakly exogenous, an originator of shocks rather than a responder. We estimate the model by maximum likelihood with a restricted constant inside the cointegration space.

3.5 Bayesian cross-check

Because point estimates hide their own uncertainty, we re-estimate the core transmission relationship in a Bayesian frame. On stationary daily log returns we specify

$$r_t^{\mathrm{SBN}} = \gamma + \beta_{\mathrm{jgb}}\, r_t^{\mathrm{JGB}} + \epsilon_t, \qquad \beta_{\mathrm{jgb}} \sim \mathcal{N}(0,1),\ \gamma \sim \mathcal{N}(0,10),\ \sigma \sim \text{Half-}\mathcal{N}(1), \tag{2}$$

and sample the posterior with the No-U-Turn Sampler (1,000 tuning and 1,000 draws, target acceptance 0.9). We report the posterior mean of β_{jgb} and the share of its mass above zero, following Gelman et al. (2013). The weakly informative prior keeps the estimate honest on a short, noisy sample without forcing a sign.

3.6 Monte Carlo

To characterize the tails rather than the average, we run 1,000 Monte Carlo paths over a 30-day horizon. Each path draws daily innovations from a multivariate normal with covariance equal to the estimated VECM residual covariance $\hat{\boldsymbol{\Sigma}}_u$ and accumulates them onto the last observed level. The output is a fan of trajectories whose central tendency we read against its dispersion. Because the residual covariance preserves the cross-equation correlation, a stress draw on the JGB equation propagates into the rupiah equation through the same channel the VECM identifies.

4 Results

4.1 Co-movement and correlation

Figure 2 reports the correlation matrix of daily log returns. The pairwise correlations are modest, which is expected and important: daily returns across these markets are noisy and weakly linked, so any genuine relationship lives in the levels and the long run, not in same-day co-movement. This is the empirical reason the analysis is built on cointegration rather than a returns regression, and it is worth holding onto when reading the loading coefficients below.

4.2 Loading coefficients: who adjusts

Table 3 is the central result. The loading coefficients separate drivers from responders without ambiguity. In the JGB equation, both error-correction terms are statistically insignificant ($\alpha_{ec1} = -0.005$, $p = 0.16$; $\alpha_{ec2} = 1.14$, $p = 0.11$). The JGB does not adjust back toward equilibrium when the system is disturbed; it is weakly exogenous, which is exactly the profile of a shock originator. In the USD/IDR equation, both loadings are large and highly significant ($\alpha_{ec1} = 0.044$, $p < 0.001$; $\alpha_{ec2} = -7.28$, $p < 0.001$). The

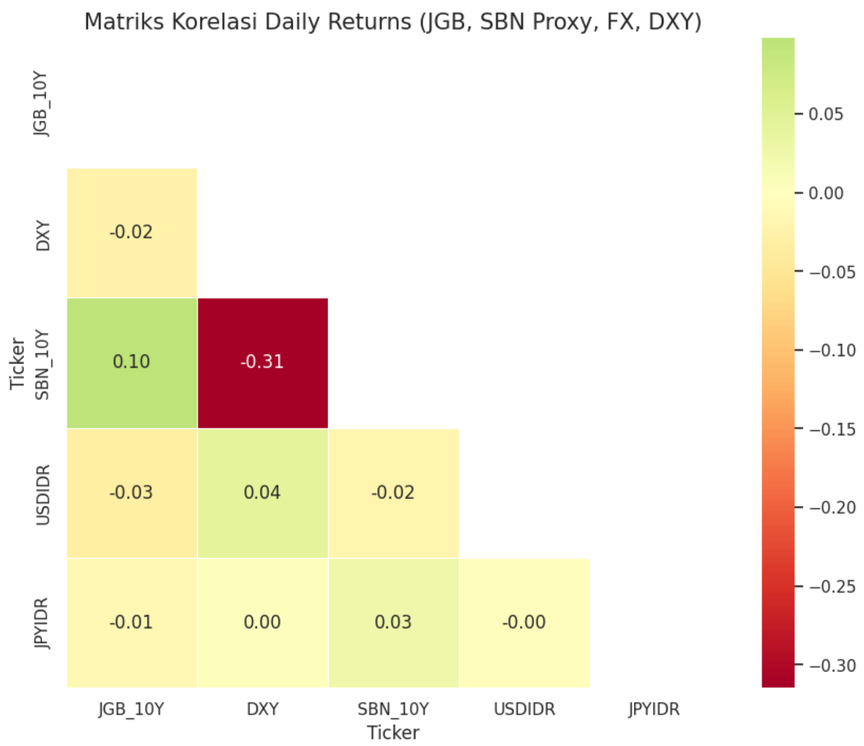

Figure 2: Correlation matrix of daily log returns. Weak same-day correlations indicate that the transmission operates through long-run equilibrium rather than contemporaneous co-movement.

rupiah carries the burden of restoring equilibrium. Read together, the two rows describe a one-directional channel: the JGB drives, the rupiah absorbs.

Table 3: VECM loading coefficients ($\boldsymbol{\alpha}$) by equation. Significance from asymptotic z-tests.

Equation	α_{ec1}	p	α_{ec2}	p
JGB proxy	-0.0054	0.159	1.1435	0.107
Indonesian proxy	-0.0001	0.010	-0.0005	0.933
DXY	0.0002	< 0.001	0.0224	0.057
USD/IDR	**0.0438**	$< \mathbf{0.001}$	$\mathbf{-7.2785}$	$< \mathbf{0.001}$

The first cointegrating vector gives the long-run relationship the rupiah is correcting toward,

$$\mathrm{JGB} + 29.70\,\mathrm{DXY} - 0.79\,\mathrm{USDIDR} = 0, \tag{3}$$

with both nonzero loadings precisely estimated (DXY coefficient 29.70, $z = 3.71$; USD/IDR coefficient -0.79, $z = -14.29$). Equation (3) says that the JGB, the dollar, and the rupiah are pinned together over the long run, and that when the combination drifts away from zero it is USD/IDR that moves to close the gap. The short-run dynamics are consistent with this: in the JGB equation the lagged Indonesian proxy enters strongly (22.91, $p < 0.001$), but the reverse channels are weak, reinforcing the direction of travel.

4.3 Impulse response

Figure 3 traces the orthogonalized response of the Indonesian proxy to a one-standard-deviation JGB shock over ten trading days. The response is not a flat line, which is the minimum bar for claiming transmission, and it does not snap back within the horizon. A disturbance in Tokyo registers in the Indonesian series and lingers, consistent with the slow error-correction the loadings imply.

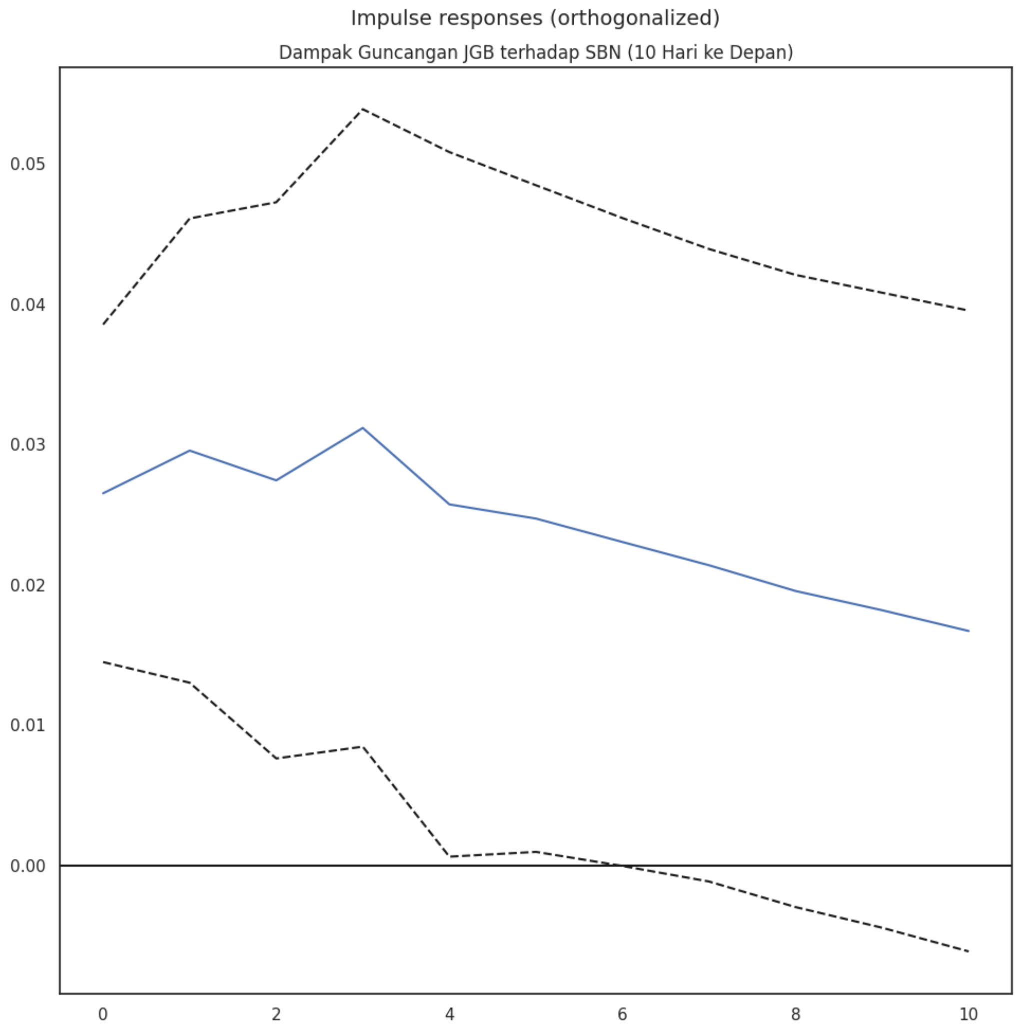

Figure 3: Orthogonalized impulse response of the Indonesian market proxy to a one-standard-deviation JGB shock, ten-day horizon. The persistence of the response is consistent with the error-correction dynamics.

4.4 Bayesian posterior

The Bayesian regression corroborates the direction with explicit uncertainty. Figure 4 shows the posterior distribution of β_{jgb}. The mass is concentrated on the positive side, with an effect size of roughly 0.16 and about 94% of the credible interval above zero. In plain terms, given the data and a neutral prior, there is roughly a fourteen-in-fifteen chance that a JGB shock pushes the Indonesian proxy in the same direction. The effect is not enormous, and the short sample keeps the posterior wide, but the sign is stable and the credibility is high enough to take seriously.

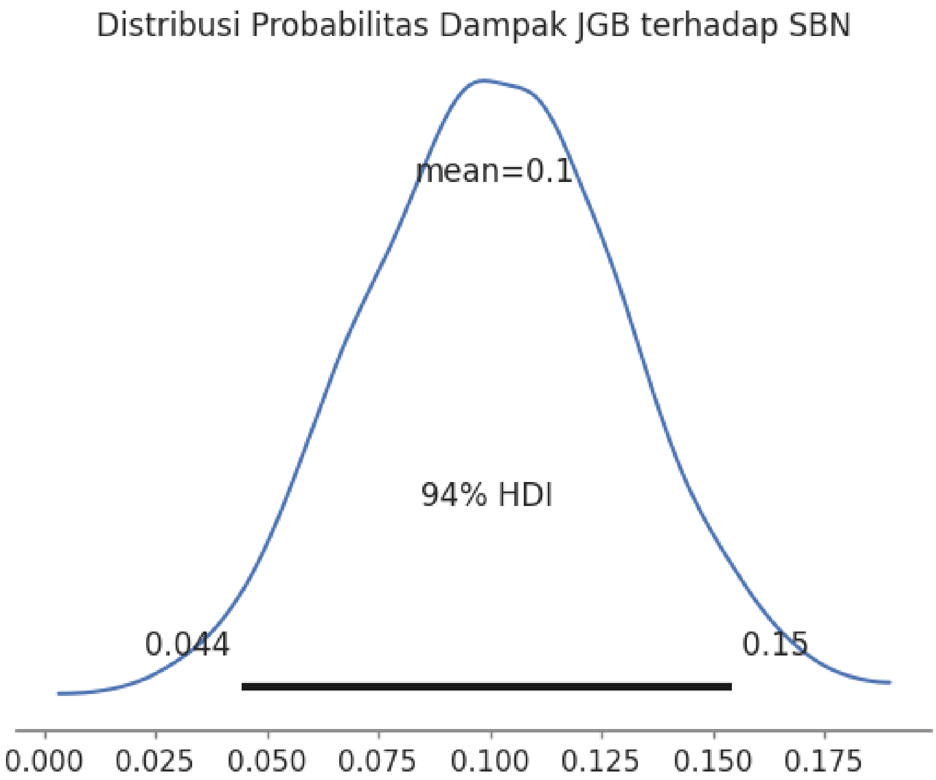

Figure 4: Posterior distribution of β_{jgb}, the transmission coefficient from JGB returns to the Indonesian proxy. Approximately 94% of the posterior mass lies above zero; posterior effect size ≈ 0.16.

4.5 Monte Carlo and the tail

Figure 5 shows the 30-day Monte Carlo fan. The central path is stable, which on its own would be a reassuring result. The story is in the spread. The trajectories fan out into heavy tails, and the dispersion is wider than a thin-tailed model would produce. This is not an artifact of the simulation: the underlying daily returns are themselves severely leptokurtic. A sensitivity regression of USD/IDR returns on DXY and JPY/IDR returns explains almost nothing ($R^2 = 0.002$) yet carries a Jarque–Bera statistic above 52,000 and an excess kurtosis near 34. The rupiah's daily distribution is dominated by rare, large moves, which is precisely the signature of a currency exposed to carry unwinds.

Translating the tail of the simulation through the cointegrating relation in Equation (3), a stressed JGB draw maps into a rupiah overshoot. With USD/IDR trading near 16,900 at the end of the sample, the stressed paths reach a band of 17,500–18,000. We flagged this in January 2026 not as a forecast of the mean but as a quantification of how bad the realistic bad case could be.

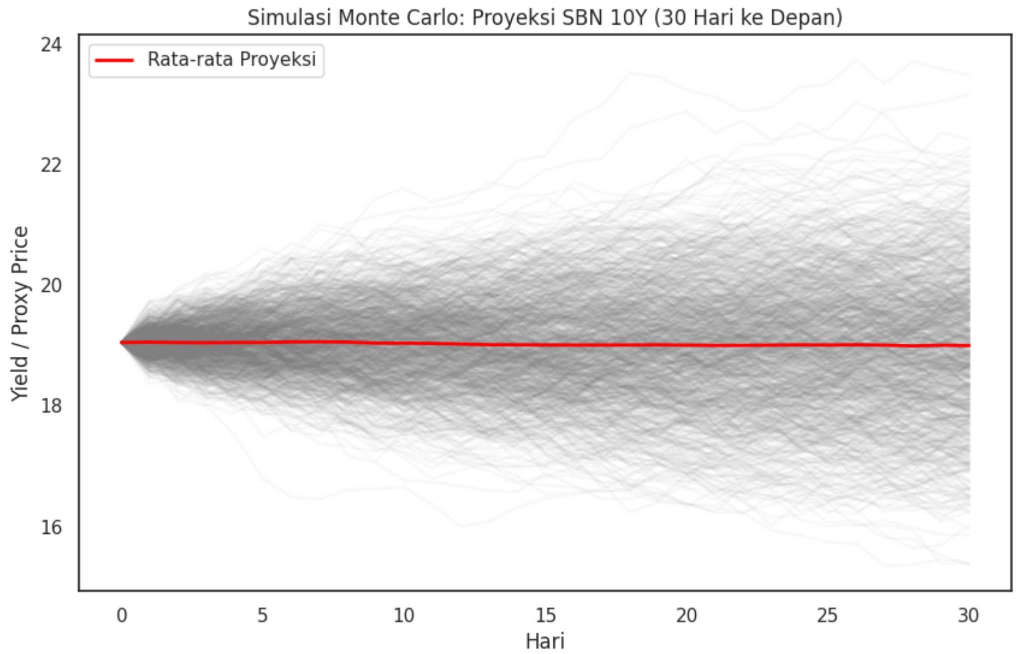

Figure 5: Monte Carlo fan chart, 1,000 simulated 30-day paths driven by the VECM residual covariance. The central path is stable while the tails are heavy, implying a stressed USD/IDR overshoot band of 17,500–18,000.

5 Discussion

The mechanism that ties these results together is the carry unwind. A higher JGB yield compresses the spread that makes the yen-funded trade worthwhile. Positions are trimmed, yen is repurchased, and the emerging assets bought with that yen are sold into whatever liquidity is available. The rupiah is among the most liquid of those assets, so it absorbs a disproportionate share of the adjustment. The econometrics line up with that narrative point for point: the JGB does not error-correct because it is the source, the rupiah error-corrects hard because it is the sink, and the dollar sits in the long-run relation as the global pricing factor that scales the whole thing.

For a policymaker the useful object is not the average path but the timing of the tail. The error-correction speed in the USD/IDR equation is fast enough that the bulk of the adjustment happens over days, not weeks. That compresses the intervention window. If BI waits for confirmation that a JGB shock has become a rupiah problem, much of the move will already have happened. The implication is that the JGB yield itself, not the rupiah, is the variable to watch, because the system gives advance notice on the driver before it shows up in the responder. An intervention rule keyed to the cointegrating gap in Equation (3) would act earlier than one keyed to the exchange rate alone.

Two cautions temper this. First, the proxies measure stress, not yields, so the magnitudes should be read as relative, not literal. Second, Forbes & Rigobon (2002) remind us that correlation inflates mechanically in volatile periods, so part of what looks like transmission during a stress episode is ordinary interdependence wearing a louder coat. Neither caution overturns the direction of the result, but both argue against over-precision.

6 Ex-Post Validation

This paper was written in January 2026, and its central tail scenario was an explicit, dated prediction: under a JGB-driven carry unwind, USD/IDR could overshoot into the 17,500–18,000 band. At the time the rupiah was trading near 16,900 and that band looked like a fat-tail caveat rather than a base case. By June 2026 the rupiah had crossed 18,000. The scenario we attached a low probability and a heavy tail to was realized within five months of the analysis.

We do not present this as a vindication of a forecast, because the Monte Carlo did not forecast a mean of 18,000; it quantified a tail. What the realization validates is narrower and, we think, more useful: that the transmission channel identified by the loadings was real, that the tail risk it implied was not a numerical artifact, and that a model flagging the danger ahead of time would have given a policymaker a usable warning. A fat tail that arrives is still a fat tail. The value of the exercise was in naming the magnitude before the fact.

7 Conclusion and Limitations

We set out to measure whether JGB shocks transmit into Indonesian markets and, if so, which variable bears the adjustment. The answer is consistent across three methods. The VECM shows a one-directional channel in which the JGB drives and the rupiah error-corrects, with loadings of 0.044 and -7.28 in the USD/IDR equation against statistically insignificant loadings on the JGB. The Bayesian regression puts the transmission coefficient at roughly 0.16 with 94% credibility on the correct side of zero. The Monte Carlo confirms that the danger is in the tail, and the tail it described in January 2026 materialized by June.

The limitations are real and we would rather state them than bury them. The ETF proxies capture risk sentiment but introduce measurement error in the direct yield channel, so the coefficients describe stress transmission rather than basis-point pass-through; replication with direct yield series from FRED and DJPPR is the obvious next step. The sample is short by macro-finance standards, which is why the Bayesian posterior is wide and why we lean on the sign rather than the magnitude. The Monte Carlo assumes Gaussian innovations even though the realized returns are far from Gaussian, which means our simulation, if anything, understates the tail. And the second cointegrating relation is weak, so the rank-two specification should be read as the more generous of two defensible choices. None of these reverse the central finding. They do argue for treating the rupiah overshoot band as a directional warning rather than a calibrated number, which is exactly how it was used, and exactly why it was worth writing down.

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